



Cloud Native Observability

Prometheus Node Exporter

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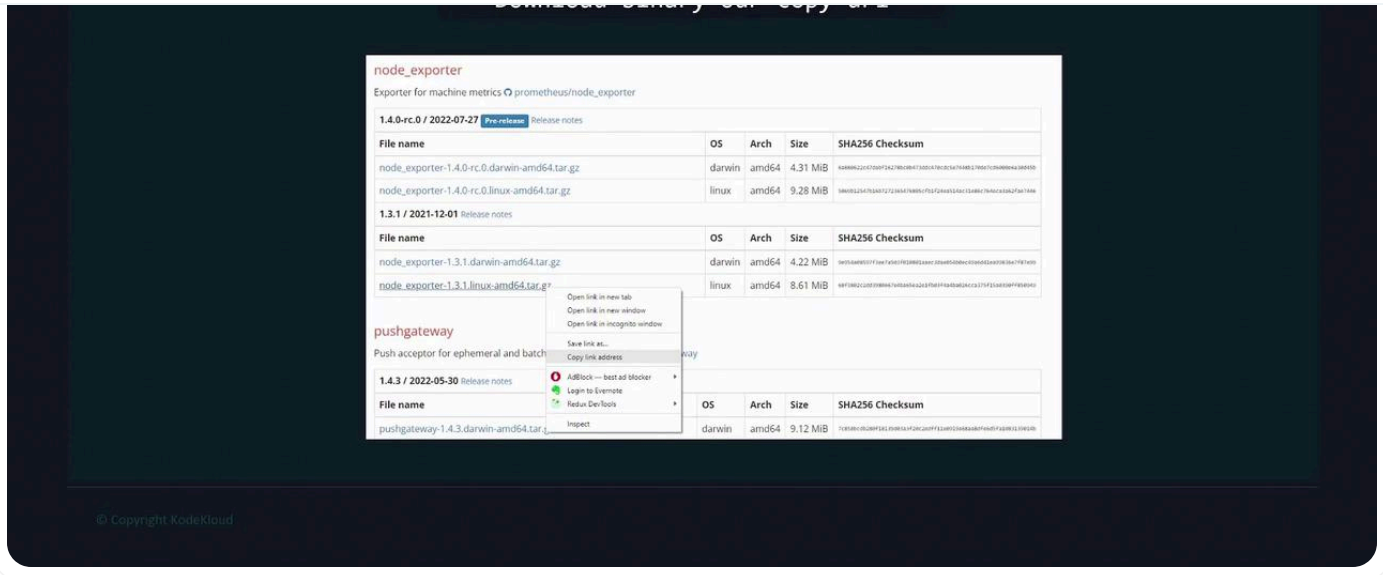
This guide explains how to install the Prometheus Node Exporter on a Linux host for system-level metrics collection.

This guide details how to set up the Prometheus Node Exporter on a Linux host. The Node Exporter collects system-level metrics and provides them in a format that Prometheus can scrape for monitoring.

Step 1: Downloading the Node Exporter

Begin by visiting the official [Prometheus downloads page](#) and selecting the Node Exporter. Choose the desired version and either download the binary directly or copy the URL for use with `wget`.

>



Once you have the URL, download the tarred file using wget. For example:

```
wget https://github.com/prometheus/node_exporter/releases/download
```



The output should resemble:

```
$ wget https://github.com/prometheus/node_exporter/releases/downlc
```



```
HTTP request sent, awaiting response... 200 OK
```

```
Length: 9033415 (8.6M) [application/octet-stream]
```

```
Saving to: 'node_exporter-1.3.1.linux-amd64.tar.gz'
```

```
node_exporter-1.3.1.linux-amd64.tar.gz 100%[=====
```

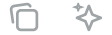
```
2022-09-02 15:04:10 (12.4 MB/s) - 'node_exporter-1.3.1.linux-amd64.tar.gz' saved
```

Verify the file's integrity by comparing its SHA256 checksum with the one provided on the download page.



>

```
tar -xvf node_exporter-1.3.1.linux-amd64.tar.gz
```



This command creates a directory named `node_exporter-1.3.1.linux-amd64` containing the executable and additional files (LICENSE and NOTICE). Change to the directory with:

```
cd node_exporter-1.3.1.linux-amd64
```



Step 3: Running the Node Exporter

Inside the directory, start the Node Exporter by running its executable:

```
./node_exporter
```



You should see output indicating that the exporter is listening on the default port 9100:

```
ts=2022-09-05T16:51:59.947Z caller=node_exporter.go:115 level=info  
ts=2022-09-05T16:51:59.947Z caller=node_exporter.go:199 level=info msg="listening  
ts=2022-09-05T16:51:59.947Z caller=tls_config.go:195 level=info msg="TLS is disabled"
```



Make sure that port 9100 is open in your firewall settings to allow Prometheus to scrape the metrics.

Step 4: Verifying the Installation

To confirm that the Node Exporter is running correctly, use curl to request the metrics endpoint:



>

```
# TYPE promhttp_metric_handler_requests_in_flight gauge
promhttp_metric_handler_requests_in_flight 1
# HELP promhttp_metric_handler_requests_total Total number of scrapes by HTTP status code
# TYPE promhttp_metric_handler_requests_total counter
promhttp_metric_handler_requests_total{code="200"} 0
promhttp_metric_handler_requests_total{code="500"} 0
promhttp_metric_handler_requests_total{code="503"} 0
```

Alternatively, open your web browser and navigate to:

<http://localhost:9100/metrics>

Example with a Different Version

The process remains the same if you are using a different version, such as `node_exporter-1.4.0.linux-amd64`. Your working directory may resemble:

```
user2 in ~
> ls
Desktop  Downloads  node_exporter-1.4.0.linux-amd64  Pictures  pushgateway
```

Change into the specific directory:

```
cd node_exporter-1.4.0.linux-amd64
```

Verify the contents:

```
ls -la
```



>

```
drwxr-xr-x 17 user2 user2    4096 Nov 12 23:43 ..
-rw-r--r--  1 user2 user2    11357 Sep 26 20:39 LICENSE
-rwxr-xr-x  1 user2 19640886 Sep 26 08:33 node_exporter
-rw-r--r--  1 user2 user2      463 Sep 26 08:39 NOTICE
```

Run the exporter:

```
./node_exporter
```



Verify the metrics with:

```
curl localhost:9100/metrics
```



A sample output provides various system metrics, including CPU, memory, and network statistics:

```
# TYPE node_timex_tick_seconds gauge
node_timex_tick_seconds 0.01
# HELP node_udp_queues Number of allocated memory in the kernel for UDP datagrams
# TYPE node_udp_queues gauge
node_udp_queues{ip="v4",queue="rx"} 0
node_udp_queues{ip="v4",queue="tx"} 0
node_udp_queues{ip="v6",queue="rx"} 0
node_udp_queues{ip="v6",queue="tx"} 0
# HELP node_uname_info Labeled system information as provided by the uname system
node_uname_info{domainname="(none)",machine="x86_64",nodename="user2",release="5.10.0-11-amd64",sysname="Linux",version="Linux version 5.10.0-11-amd64 (root@buildd.debian.org)"} 1
...
```



Conclusion



>

For further customization and enhancements, explore additional Prometheus exporters and check out the [Prometheus Documentation](#).



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